

Simulation of String Oscillations using the Velocity Verlet Method

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1 Introduction

The objective of this project is to simulate the dynamics of a one-dimensional massive string. The physical system is governed by the wave equation, augmented with terms for frictional damping and external driving forces. The vertical displacement of the string $u(x, t)$ evolves according to the partial differential equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} - \gamma \frac{\partial u}{\partial t} + F_{ext} \sin(\Omega t) \delta(x - x_p) \quad (1)$$

where $c = \sqrt{T/\rho}$ is the speed of sound (with tension T and linear mass density ρ), γ is the damping coefficient, and the last term represents a periodic external force applied at a specific point x_p .

We investigate various scenarios, including reflection at boundaries, propagation of wave packets, impedance mismatch in non-homogeneous media, spectral decomposition of modes, and mechanical resonance.

2 Numerical Method

2.1 Mesh, finite difference method, discretization

To simulate the wave dynamics numerically, the continuous string of length L is approximated by a discrete grid of points. We divide the spatial domain into N equidistant nodes, resulting in a spatial resolution defined by:

$$\Delta x = \frac{L}{N-1}, \quad x_i = \Delta x \cdot i, \quad i = 0, 1, 2, \dots, N-1 \quad (2)$$

Similarly, the time evolution is discretized into steps of size Δt :

$$t_n = \Delta t \cdot n, \quad n = 0, 1, 2, \dots \quad (3)$$

To ensure numerical stability, the time step is coupled to the spatial step via the Courant-Friedrichs-Lewy (CFL) condition. We utilize a safety factor $\alpha \leq 1$ (specifically $\alpha = 0.5$) to define:

$$\Delta t = \alpha \frac{\Delta x}{c} \quad (4)$$

Using the notation u_i^n for displacement and v_i^n for velocity at node i and time n , the discrete form of the driven, damped wave equation is given by:

$$a_i^n = c_i^2 \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} - \gamma v_i^n + F_{ext} \sin(\Omega t_n) \delta_{i,i_p} \quad (5)$$

2.2 Boundary conditions (BC)

The behavior of the wave at the string's extremities is governed by boundary conditions. We investigate two distinct physical scenarios:

- *Fixed BC*: Corresponds to a Dirichlet condition where the endpoints are clamped and cannot move.

$$u(0, t) = u(L, t) = 0 \implies u_0^n = u_{N-1}^n = 0 \quad (6)$$

- *Open BC*: Corresponds to a Neumann condition where the derivative is zero, implying the ends are free to slide transversely.

$$\left. \frac{\partial u}{\partial x} \right|_{x=0, L} = 0 \implies u_0^n \leftarrow u_1^n \text{ and } u_{N-1}^n \leftarrow u_{N-2}^n \quad (7)$$

Note that for the open boundary implementation, we update the ghost nodes based on their neighbors, and explicit velocity calculations at these boundaries are not required for the algorithm.

2.3 Kinetic and potential energies

In a lossless system, the total mechanical energy is an invariant of motion. Monitoring this quantity serves as a crucial validation of the numerical scheme's stability. The total kinetic energy is computed by summing the contributions of all moving internal mass elements:

$$E_{kin}(t_n) = \sum_{i=1}^{N-2} \frac{m_i (v_i^n)^2}{2} = \sum_{i=1}^{N-2} \frac{\Delta x \rho_i (v_i^n)^2}{2} \quad (8)$$

The potential energy arises from the elastic stretching of the string segments. Using a backward difference approximation for the gradient, we integrate the tension work along the string:

$$E_{pot}(t_n) = \sum_{i=1}^{N-1} \frac{\Delta x T}{2} \left(\left. \frac{\partial u}{\partial x} \right|_{x=x_i} \right)^2 = \sum_{i=1}^{N-1} \frac{\Delta x T}{2} \left(\frac{u_i^n - u_{i-1}^n}{\Delta x} \right)^2 \quad (9)$$

As noted in the reference method, the summation indices for potential energy run from 1 to $N - 1$ due to the spatial derivative stencil used.

2.4 Velocity Verlet method

We employ the Velocity Verlet integration scheme, a symplectic algorithm favoured for its energy conservation properties in oscillatory systems. The state update from step t to $t + \Delta t$ proceeds as follows:

1. **Time update:** Increment simulation time.
2. $t \leftarrow (it - 1) \cdot \Delta t$
3. **Initial Acceleration:** Compute forces based on current configuration.
4. $a_i \leftarrow c_i^2 \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} - \gamma \cdot v_i + F_{ext} \sin(\Omega t) \cdot \delta_{i, i_p}, \quad i = 1, 2, \dots, N - 2$
5. **Half-step Velocity:** Predict velocity at the midpoint $t + \Delta t/2$.

$$\tilde{v}_i \leftarrow v_i + \frac{\Delta t}{2} a_i, \quad i = 1, 2, \dots, N - 2 \quad (10)$$

6. **Position Update:** Advance positions using the half-step velocity.

$$u_i \leftarrow u_i + \tilde{v}_i \Delta t \quad (11)$$

7. **Apply Boundary Conditions:** Enforce constraints on the new positions.

- $u_0 \leftarrow 0$ and $u_{N-1} \leftarrow 0$ for *fixed BC*
- OR
- $u_0 \leftarrow u_1$ and $u_{N-1} \leftarrow u_{N-2}$ for *open BC*

8. **New Acceleration:** Recalculate forces at $t + \Delta t$ using the new displacements u and the half-step velocities $\tilde{v}$ (for the damping term).

$$a_i \leftarrow c_i^2 \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} - \gamma \cdot \tilde{v}_i + F_{ext} \sin(\Omega(t + \Delta t)) \cdot \delta_{i,i_p}, \quad i = 1, 2, \dots, N-2 \quad (12)$$

9. **Full-step Velocity:** Correct the velocity to the end of the step.

$$v_i \leftarrow \tilde{v}_i + a_i \frac{\Delta t}{2}, \quad i = 1, 2, \dots, N-2 \quad (13)$$

10. **Output:** Store t, E_{kin}, E_{pot} and displacement profile u_i .

3 Results

3.1 Task 1: Fixed Boundary Conditions

A Gaussian pulse is initialized at the center of a homogeneous string with fixed ends.

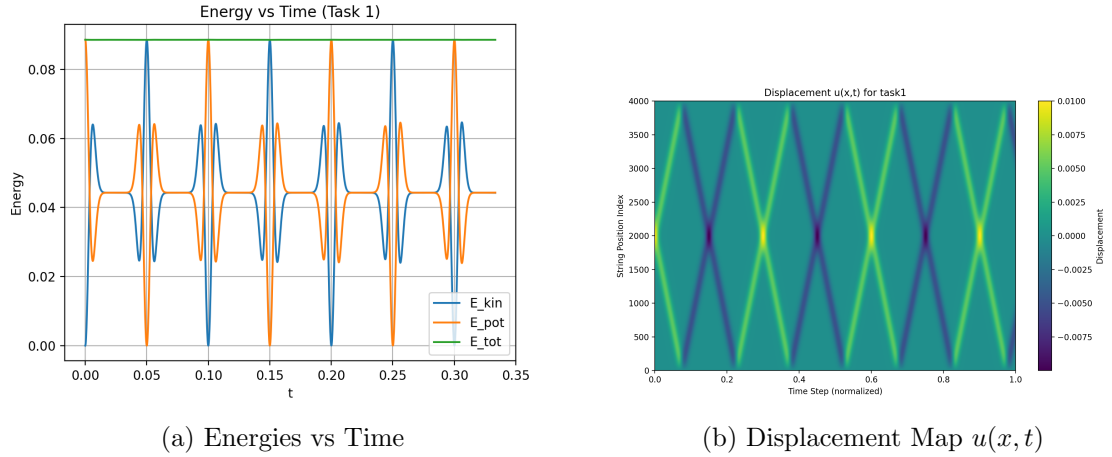
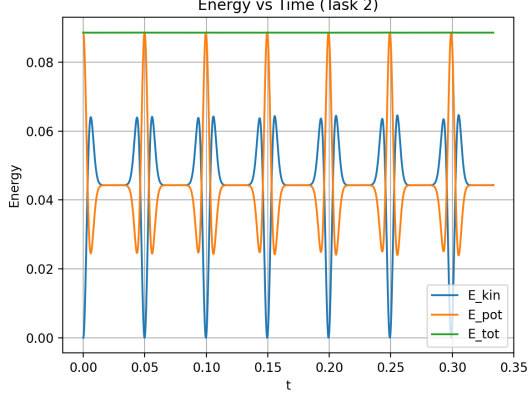


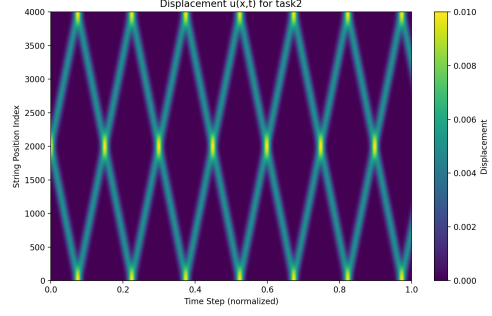
Figure 1: Task 1: The wave packet splits into two counter-propagating waves. Upon reaching the fixed boundaries, the waves reflect with an inversion of amplitude (phase shift of π), visible in the blue/red alternation in the heatmap.

3.2 Task 2: Open Boundary Conditions

The simulation is repeated with open (Neumann) boundary conditions.



(a) Energies vs Time



(b) Displacement Map $u(x,t)$

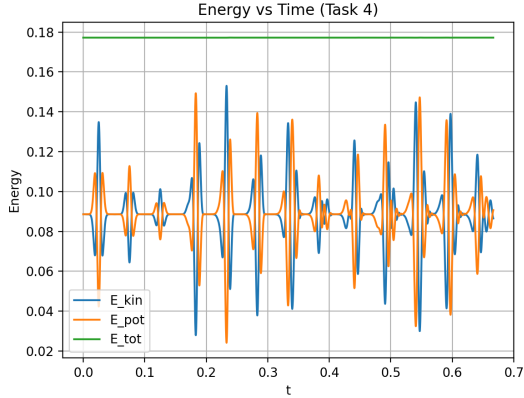
Figure 2: Task 2: Unlike the fixed case, reflection at open boundaries occurs without phase inversion. The wave maintains its sign after reflection. Total energy remains conserved (constant) in both Task 1 and 2, validating the symplectic nature of the Verlet method.

3.3 Task 3: Moving Wave Packet

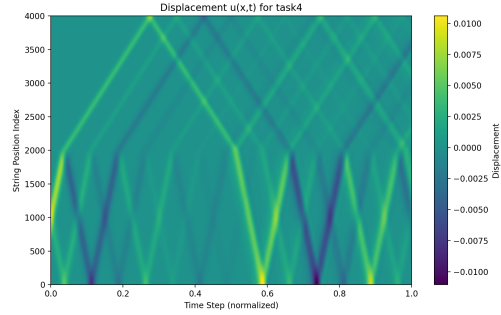
We investigate the effect of initial velocity v_{init} on wave propagation.

3.4 Task 4: Non-homogeneous String

The string density changes at $x = L/2$: $\rho(x > L/2) = 10\rho_0$.



(a) Energies vs Time



(b) Displacement Map

Figure 3: Task 4: An impedance mismatch occurs at the interface. Part of the wave is transmitted into the denser medium (propagating slower, as seen by the steeper slope in the space-time plot), and part is reflected. Energy is conserved.

3.5 Task 5: Spectral Decomposition and Damping

Friction ($\gamma = 150$) is introduced. We analyze the decay of the first three fundamental modes.

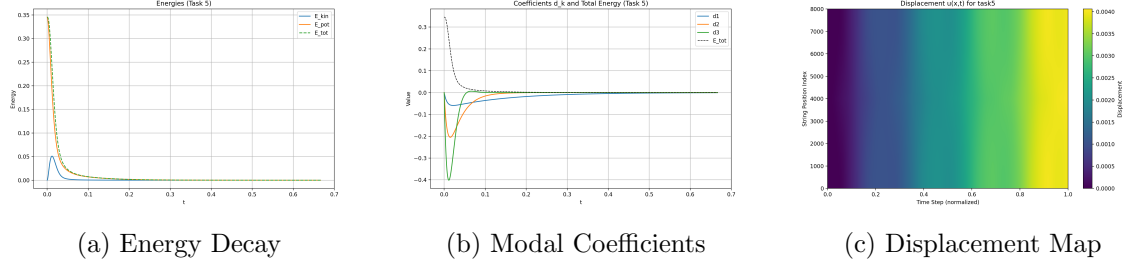


Figure 4: Task 5: The total energy decreases over time due to damping. Higher frequency modes decay faster than lower frequency modes, which is consistent with the physics of damped oscillations.

3.6 Task 6: Resonance

The string is driven by an external force $F_{ext} \sin(\Omega t)$ at point x_p .

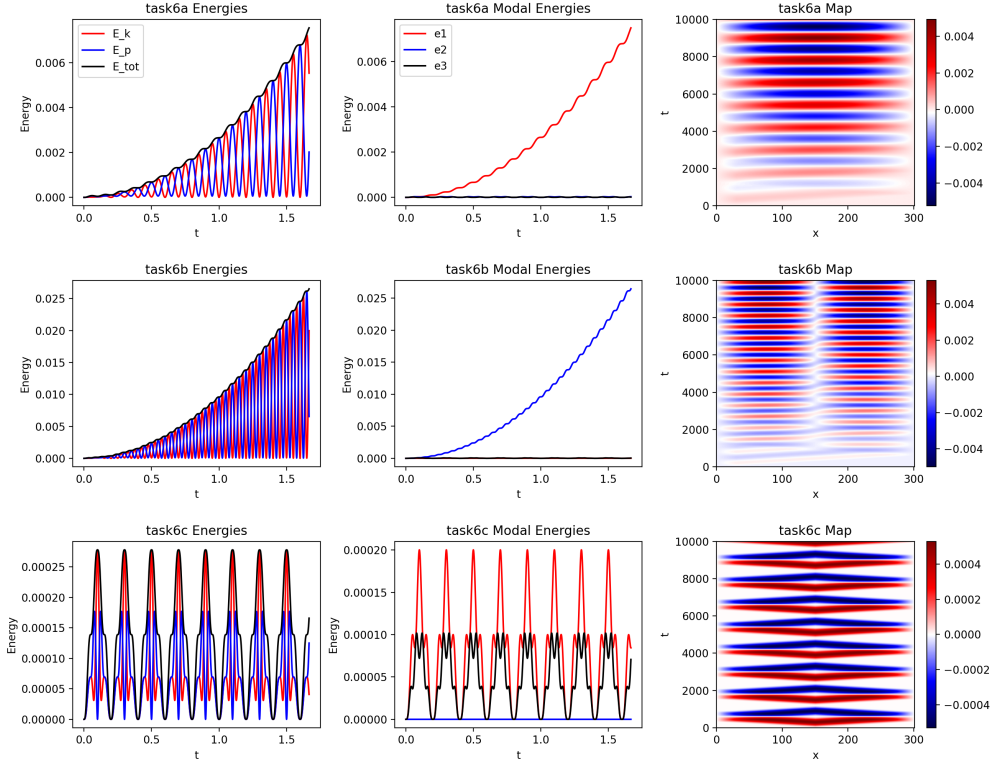


Figure 5: Task 6: Top Row (6a): Driving at $\Omega = \omega_1$, $x_p = 30$. Resonance of the 1st mode is clearly visible; energy grows quadratically. Middle Row (6b): Driving at $\Omega = \omega_2$, $x_p = 30$. Resonance of the 2nd mode. Bottom Row (6c): Driving at $\Omega = \omega_2$, $x_p = (N-1)/2$ (center). The center is a node for the 2nd mode. Consequently, the force does no work, and the 2nd mode is not excited, demonstrating that resonance requires both frequency matching and spatial coupling.

4 Conclusions

The Velocity Verlet method proved to be a robust and stable algorithm for solving the wave equation.

1. **Conservation Laws:** In the absence of damping (Tasks 1-4), the total energy was conserved to a high degree of precision, confirming the symplectic nature of the integrator.
2. **Boundary Conditions:** The distinction between Dirichlet (inversion) and Neumann (no inversion) boundaries was numerically verified.
3. **Wave Propagation:** The simulation correctly reproduced the d'Alembert solution behavior, where an arbitrary pulse splits unless the initial velocity is perfectly matched to the displacement gradient ($v = -c \frac{\partial u}{\partial x}$ for right-moving).
4. **Resonance:** The most significant observation was in Task 6. Resonance is not solely a frequency phenomenon. Even if the driving frequency matches a natural eigenfrequency ($\Omega = \omega_2$), resonance does not occur if the force is applied at a node of that standing wave mode. This highlights the importance of spatial overlap between the driving force and the mode shape.

Overall, the project successfully demonstrated fundamental wave mechanics concepts through direct numerical simulation.